Government of Uttar Pradesh

Department of Excise

उत्तर प्रदेश आबकारी विभाग

Ask the Data

Questions in plain language, answered from the department's own figures and rules, on the department's own server

Proof of concept

The problem: a simple question can take days

Figures sit in many files

Revenue, dispatch, shop and enforcement figures are held in separate workbooks, reports and portals.

Each question needs an analyst

An officer asks, an analyst finds the files, builds the table and sends it back. A new angle on the same question starts the cycle again.

Rules live in another place

Acts, rules and policy orders are published as documents. Linking a rule to a figure is done by hand.

An officer who needs a number for a meeting today should be able to get it today, with the source shown, without waiting for a report to be prepared.

What this proof of concept does

An officer types a question in everyday English. The system finds the answer in the department's data, draws a chart where it helps, and shows the exact query it used.

Ask

One question, one answer page: the figure, a chart, a plain summary and the query behind it.

Chat

A conversation that can look up figures and rules as it goes, and take follow-up questions.

Knowledge library

Excise acts, rules and policy orders, searchable in the same chat.

It runs on the departmental server, completely local. No external AI service provider and no closed-source or proprietary tool is used; all models run locally.

How an answer is made

1 You ask

Type the question the way you would put it to a colleague.

2 The system understands

An AI model turns the question into a precise search of the data.

3 The data answers

The search runs against the department's data bank, read-only.

4 You see the result

A figure, a chart, a short summary and the exact query used.

Every answer shows its working. The query is visible and can be copied, so an officer or an auditor can check where a number came from.

Live example: how many shops, and of which kind?

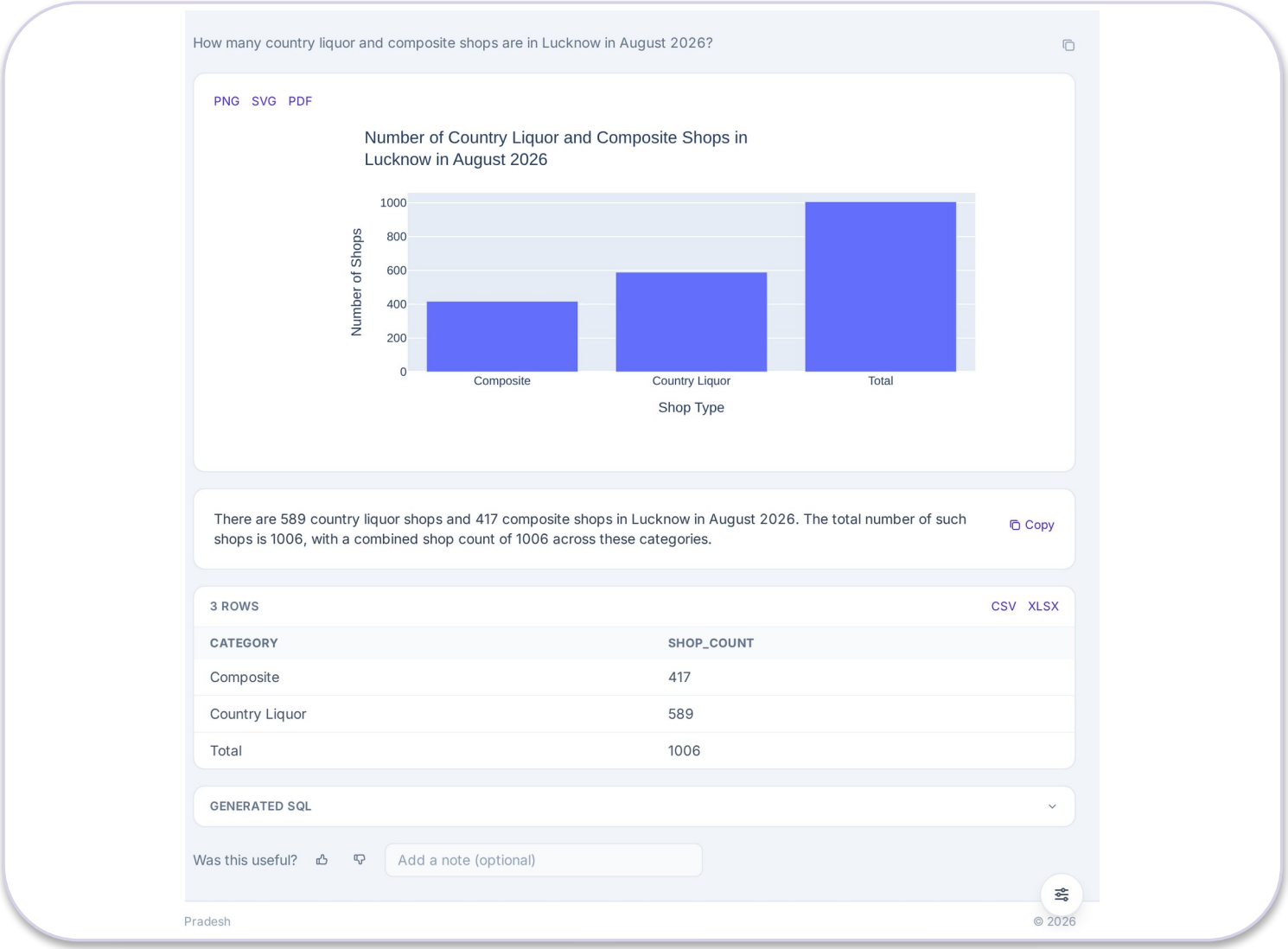
"How many country liquor and composite shops are in Lucknow in August 2026?"

Total: 1,006 shops.

Split: 589 country liquor and 417 composite.

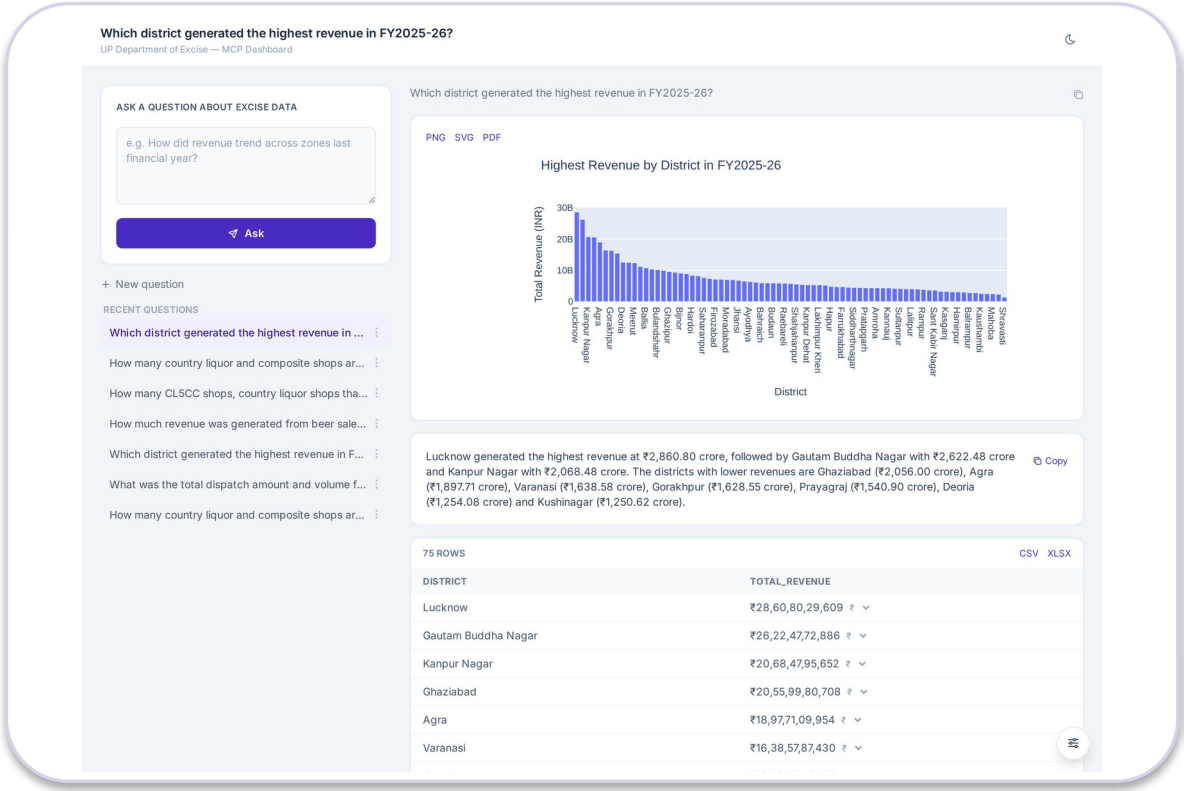
The total is counted by the database itself, not added up by the AI.

Source: the monthly dispatch report from the Integrated Excise Supply Chain Management System (IESCMS).



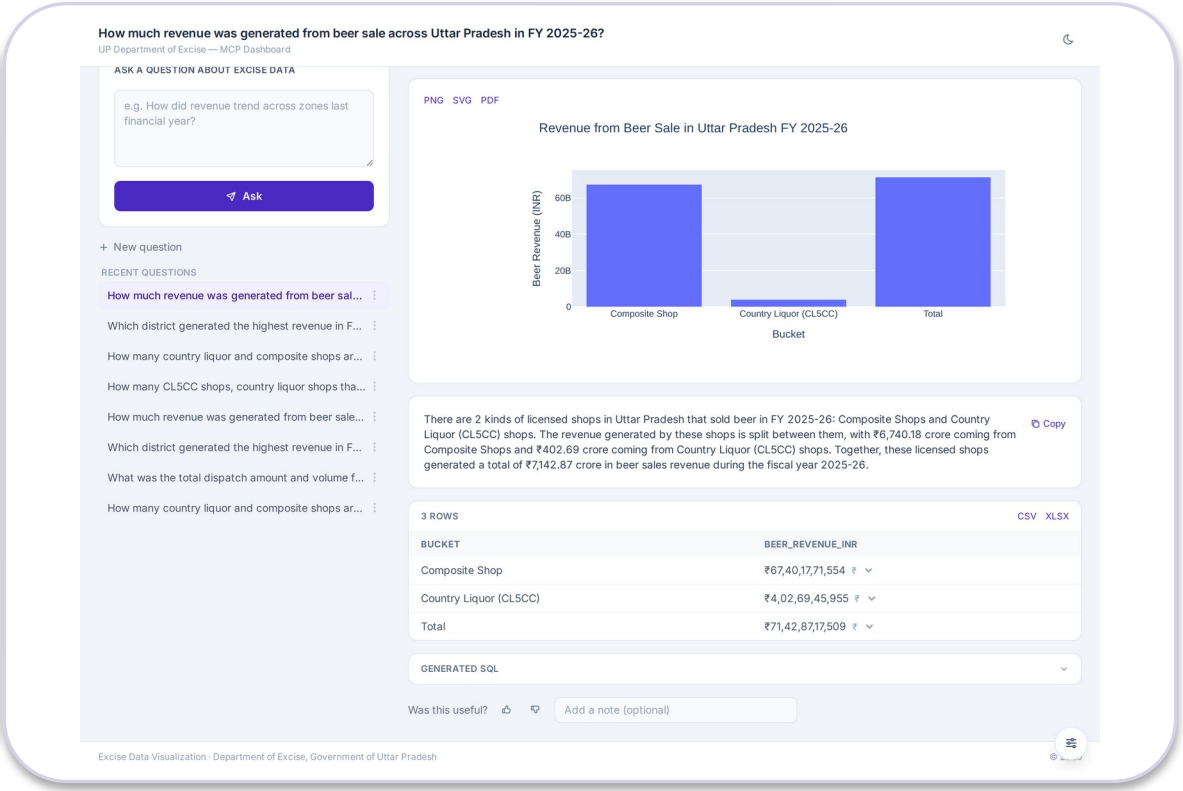
Live example: revenue by district and for a product

"Which district generated the highest revenue in FY2025-26?"



Lucknow is the highest, at ₹2,860.80 crore out of a state total of about ₹55,463 crore.

"How much revenue came from beer sale across Uttar Pradesh in FY2025-26?"



Beer revenue is ₹7,142.87 crore: ₹6,740.18 crore from composite shops and ₹402.69 crore from country liquor shops that also sell beer (CL5CC).

Live example: a conversation with a chart

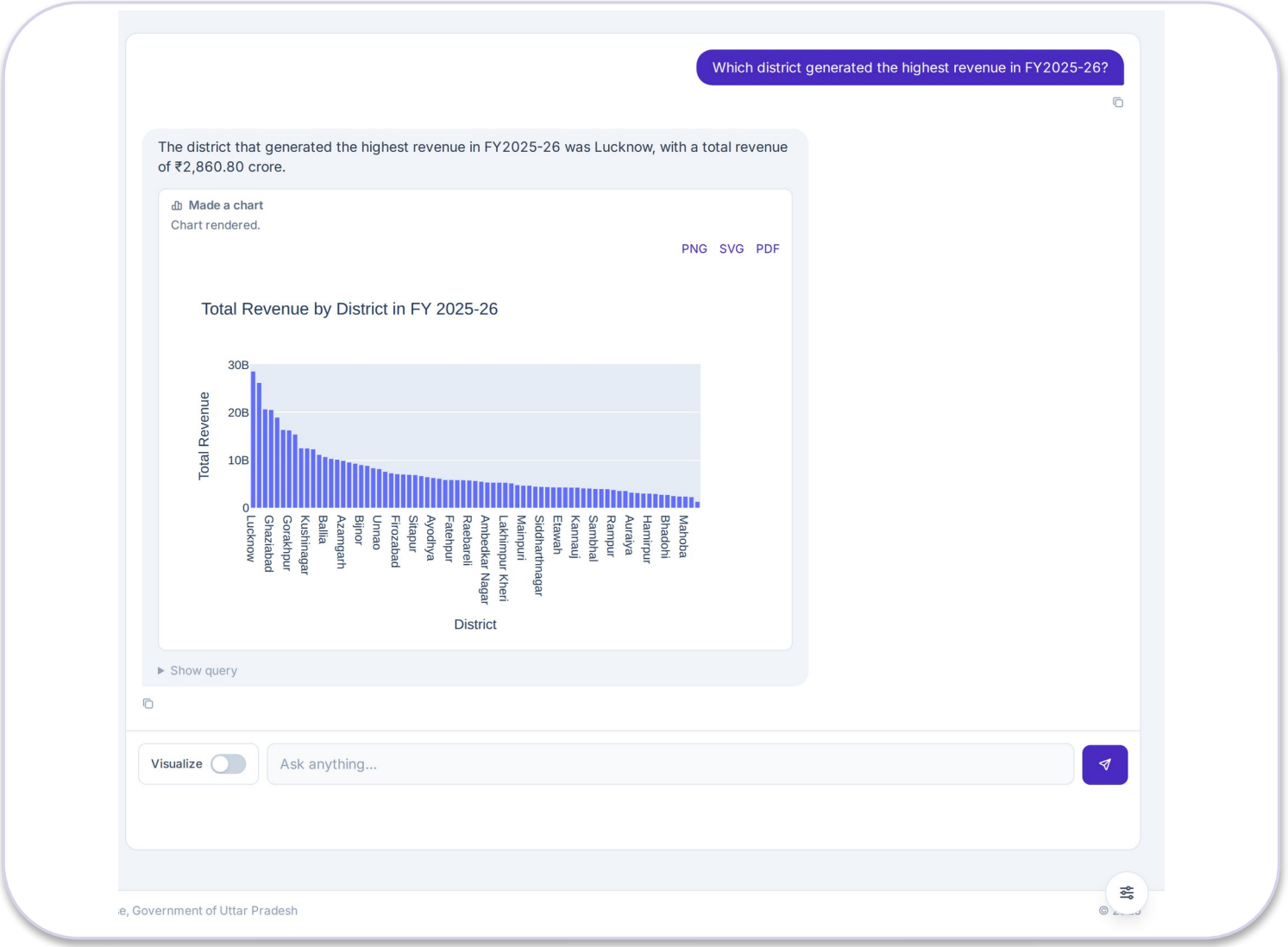
"Which district generated the highest revenue in FY2025-26?"

The same question in Chat: the answer first, then the chart.

Lucknow is highest at ₹2,860.80 crore, from all 75 districts.

The chart downloads as PNG, SVG or PDF, and the whole conversation exports as a PDF.

The query behind the answer stays one click away.



The data behind the answers

9,322

dispatch records for Lucknow, August 2026, from the Integrated Excise Supply Chain Management System

28,405

shops with FY2025-26 revenue, across all 75 districts

3,455

district revenue records from the Rajkosh and Koshwani portal data

3,812

sales-volume records by liquor type and year

11,124

enforcement records: raids, seizures, convictions

330+

acts, rules and policy documents in the knowledge library

All of it is held on the department's own server, in a database the AI can read but never change.

The AI models

Two open-source models run on the office server. Each has one job. Nothing is sent to an outside service.

Llama 3.1 (8 billion)

The conversationalist

Talks with the officer, decides when to look something up, and writes the plain-language summary.

Qwen 2.5 Coder (7 billion)

The specialist

Writes the database search and the chart code. It sees the data layout, not the officer's chat.

Why two models: a conversational model and a code-writing model are each better at their own task.

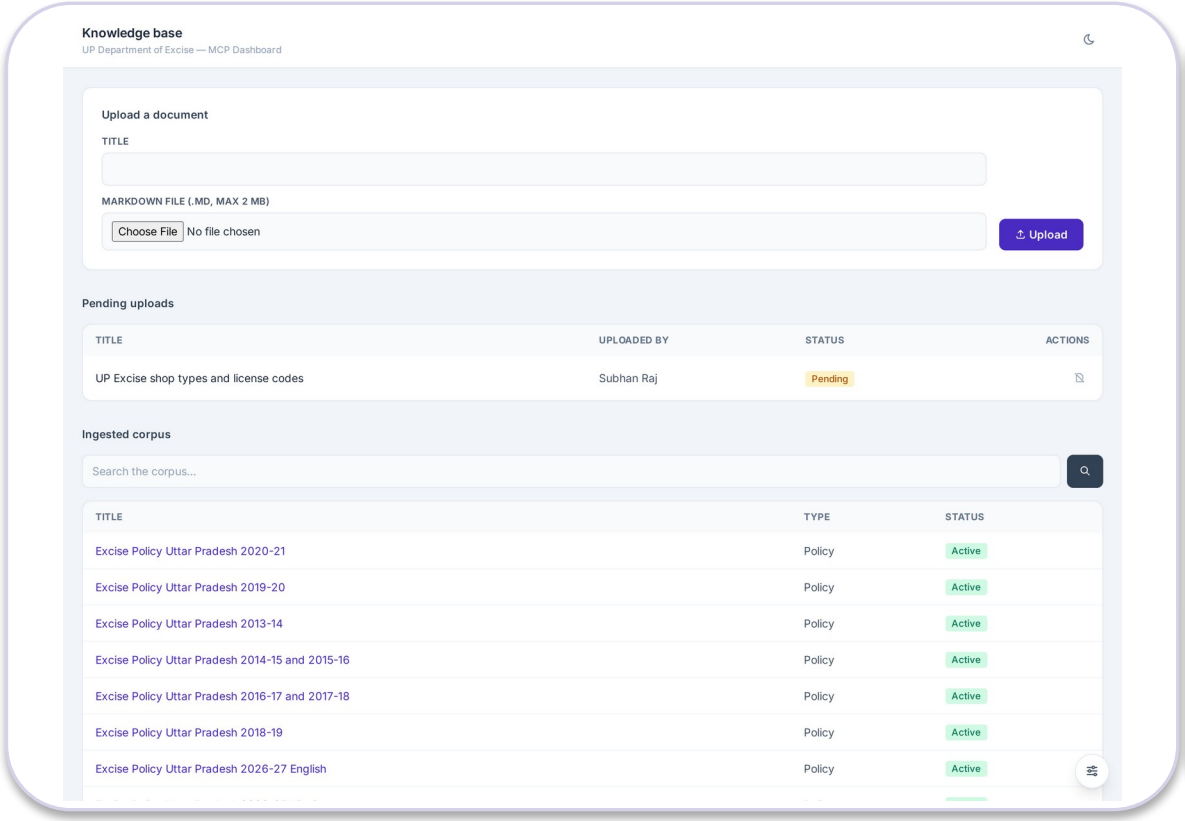
Why local: the department's questions and figures never leave the building.

Safety and control

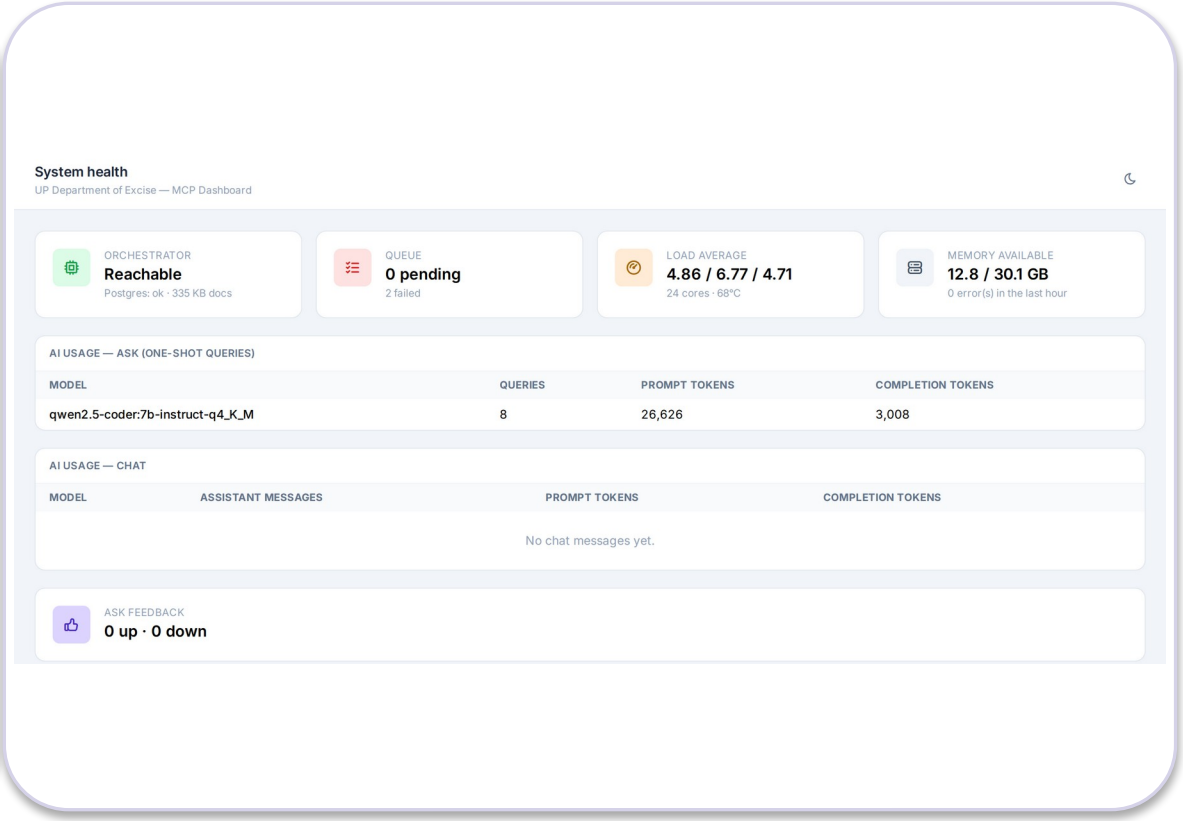
Sign-in	Email one-time code. Each person has a role: Admin or Analyst.
Read-only data	The AI connects with a database account that can read but cannot change anything.
Checked queries	Every query is inspected first. Only a single read request is allowed.
Locked box for charts	Chart code runs in an isolated space: no network, limited time and memory.
Full record	Every question, query and action is logged for audit.

The only outside connection is the secure tunnel that lets officers reach the site.

What administrators see



Knowledge base: upload a rule note, see every document the chat can quote, and withdraw one.



System health: whether the AI service is up, the queue, the server load, and how much each model was used.

How we built it, and how we tested it

Data bank

Department data loaded and cleaned. Rows that fail checks are set aside, not guessed.

Question engine

Question to query to answer, with a retry when the first attempt fails.

Knowledge library

Published rules split into searchable sections.

Chat and Ask screens

Web screens for officers and for administrators.

Real-question testing

Each demo question was run live and the answer checked against the source.

The automated tests cover the safety checks, the data loaders and the chat flow. Live testing against real questions found problems the tests could not, and each one was fixed and recorded.

What we learned, and the limits today

What we learned

A language model can give a confident wrong answer. We added checks so an empty result is reported as empty.

District revenue after 2018 in the Rajkosh and Koshwani portal data is booked at the central treasury in Lucknow. We cross-checked against shop-level revenue and now point the system to the correct source.

Money conversions are done in code. A model got one crore figure wrong by ten times.

Limits today

The server has no graphics card, so an answer takes one to three minutes.

Shop-level monthly detail is loaded for Lucknow only (August 2026).

Shop-level Bhang data has no source yet.

Answers should be treated as a fast first reading, to be checked against the source for formal use.

What comes next

High-performance compute

A high-performance computing (HPC) node with high-end GPU servers runs larger, more capable models and shortens response times.

More data

Monthly feeds for every district, and the missing sources such as Bhang shops.

Saved analyses and reports

Re-run a favourite question each month and export a report with the data date on it.

Wider access

Roles for district and division officers, with their own question history.

The decision we are asking for

This proof of concept shows that officers can get sourced answers from the department's own data in minutes, on the department's own server.

We ask the respected authorities for a decision on the next step.

Proposed next steps

1. A pilot with a small group of district and division officers.
2. A high-performance GPU compute node.
3. Monthly data feeds from the source systems and involved stakeholders.